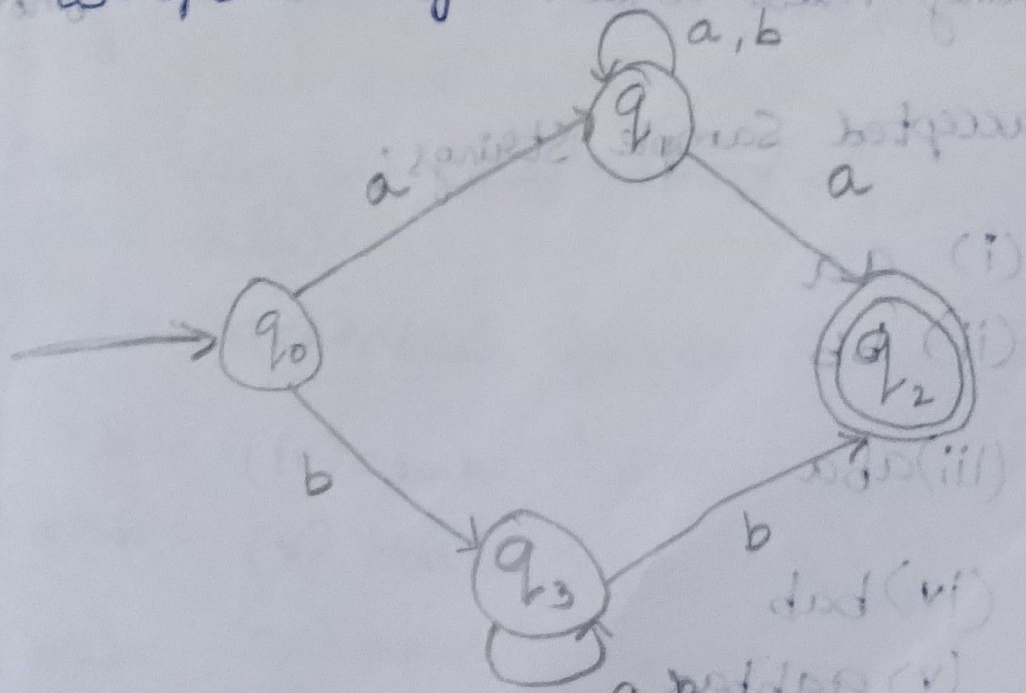
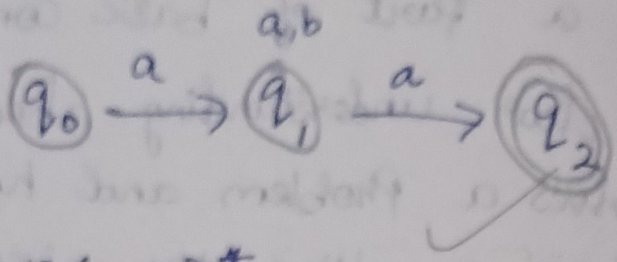
Assignment - 1

① Find the language represented by the finite automata given below and give an informal description. Write 5 Sample strings that will accepted by the automata.



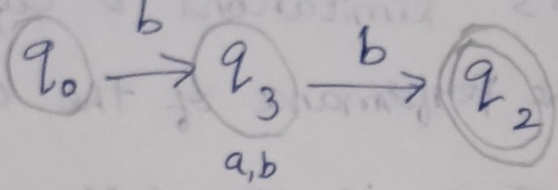
⇒ Two Possible Paths,

Path 1,



$a(a+b)^*a$

Path 2,



$b(a+b)^*b$

Combining both,

$a(a+b)^*a + b(a+b)^*b$

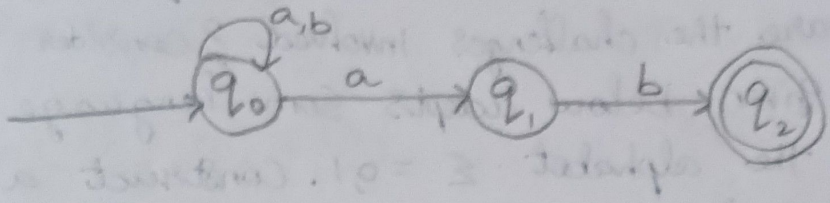
Informal description:

The language consists of all strings over {a,b} and start and end with same symbol and string must contain at least two symbols.

5 accepted sample strings:

(i) aa
(ii) bb
(iii) aba
(iv) bab
(v) aabbaa

(2) Find the language represented by finite automata given below and give an informal description. Write 10 sample strings that will be accepted by the automata



⇒ Start State : q_0

Final State : q_2

q_0 has loop on a,b

$q_0 \rightarrow q_1$ on a

$q_1 \rightarrow q_2$ on b

Language represented, $L = \{ w \in (a,b)^* \mid w \text{ ends with } ab \}$

read a $q_0 \rightarrow q_1$

immediately read b $q_1 \rightarrow q_2$

Regular expression $= (a+b)^*ab$

Informal description :

The language consists of all strings over $\{a,b\}$ that end with ab.

(There can be any number a's & b's before final ab.)

10 accepted Sample Strings :

(i) ab
(ii) aab
(iii) bab
(iv) abab
(v) bbab
(vi) aaab
(vii) babab
(viii) aabab
(ix) abbab
(x) bababab